

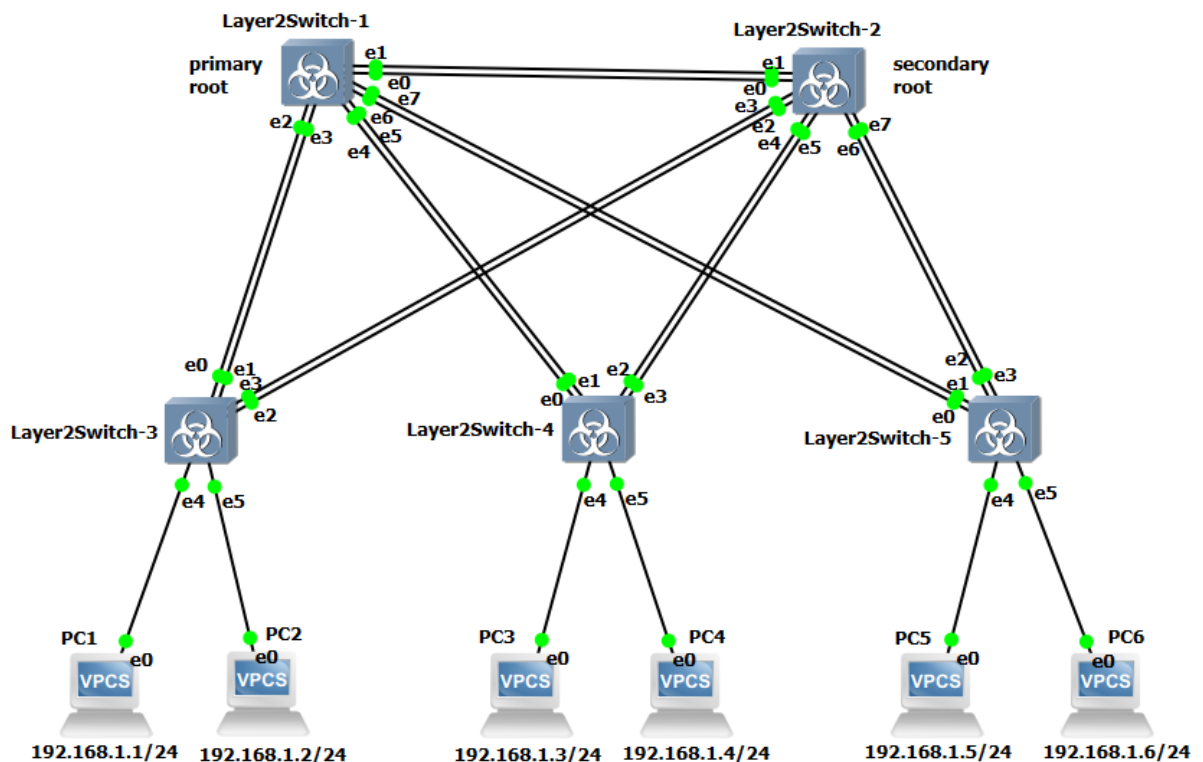
Лабораторная работа №2

Тема: Настройка протокола STP (IEEE 802.1D)

1. В заданной схеме настроить STP, назначив явно один из коммутаторов корневым настройкой приоритета;

Топология схема представлена на рисунке. Корневым является коммутатор под номером 1. Резервным корневым - коммутатор под номером 2.

2. Исходная схема также представлена в файле «lab2 1.png».



2. Проверка доступности каждого с каждым персональных компьютером;

Проверка производилась с помощью утилиты ping, журналы выполнения команд представлены в файлах с названиями:

«lab2 PC[1-6] pings.log».

3. На изображении схемы отметить BID каждого коммутатора и режимы работы портов, и стоимости маршрутов;

Изменённая схема представлена в файле «lab2 3.png».

4. Отслеживание передачи пакетов hello от корневого коммутатора;

На линках e0-e0 и e1-e1 между Switch-1 и Switch-2 корневой коммутатор отправляет hello-пакеты, MAC-адрес источника – Switch-1 (порты 0 и 1), Root BID – Switch-1 (Как нулевой порт).

Захват с Standard input [Layer2Switch-1 Ethernet1 to Layer2Switch-2 Ethernet1]

No.	Time	Source	Destination	Protocol	Length	Info
8	6.832713	0c:f9:f3:18:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8002
9	8.839767	0c:f9:f3:18:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8002
10	10.839466	0c:f9:f3:18:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8002

Захват с Standard input [Layer2Switch-1 Ethernet0 to Layer2Switch-2 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
25	30.049478	0c:f9:f3:18:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8001
26	32.049164	0c:f9:f3:18:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8001
27	33.237384	0c:f9:f3:18:00:00	CDP/VTP/DTP/PAGP/UD...	DTP	62	Dynamic Trunk Protocol
28	34.049869	0c:f9:f3:18:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8001

На линках e2-e0 и e3-e1 между Switch-1 и Switch-3 у пакетов MAC-адрес источника – Switch-1, Root BID – Switch-1.

Захват с Standard input [Layer2Switch-1 Ethernet3 to Layer2Switch-3 Ethernet1]

No.	Time	Source	Destination	Protocol	Length	Info
16	16.018509	0c:f9:f3:18:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8004
17	17.815595	0c:f9:f3:18:00:03	0c:f9:f3:18:00:03	LOOP	60	Reply
18	18.021202	0c:f9:f3:18:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8004
19	20.023910	0c:f9:f3:18:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8004

Захват с Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
17	14.467611	0c:f9:f3:18:00:02	0c:f9:f3:18:00:02	LOOP	60	Reply
18	15.214517	0c:f9:f3:18:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8003
19	17.217158	0c:f9:f3:18:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8003
20	19.216863	0c:f9:f3:18:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8003

На линках e4-e0 и e5-e0 между Switch-3 и PC1, PC2 у пакетов MAC-адрес источника – Switch-3(порты 4 и 5), Root BID – Switch-1 (порт 0).

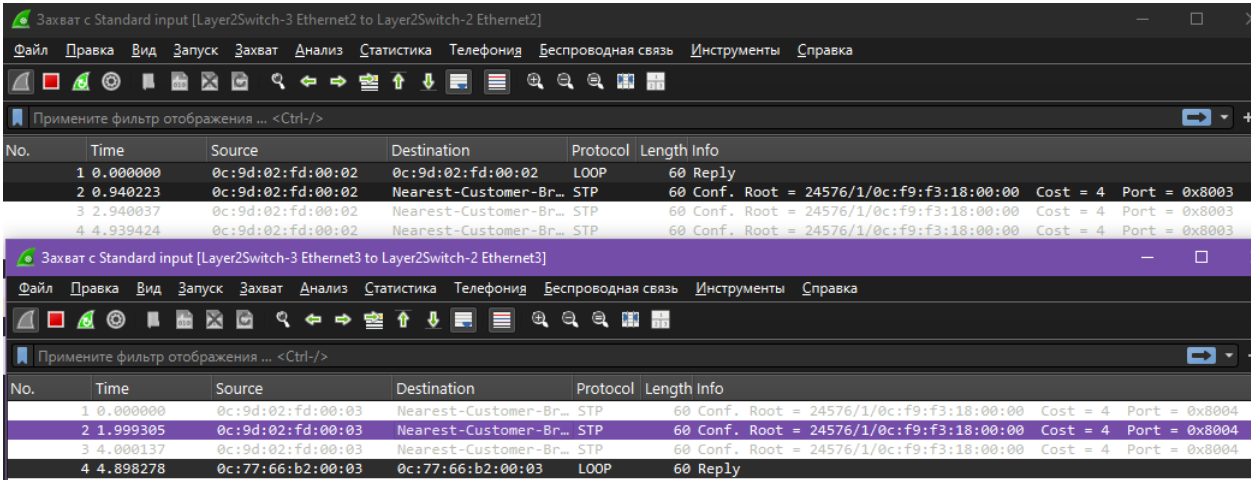
Захват с Standard input [Layer2Switch-3 Ethernet4 to PC1 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
65	86.043969	0c:f9:f3:18:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
66	88.044693	0c:f9:f3:18:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
67	90.044486	0c:f9:f3:18:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
68	91.207866	0c:f9:f3:18:00:04	0c:f9:f3:18:00:04	LOOP	60	Reply

Захват с Standard input [Layer2Switch-3 Ethernet5 to PC2 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
88	122.062564	0c:f9:f3:18:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
89	124.063538	0c:f9:f3:18:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
90	126.065020	0c:f9:f3:18:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
91	127.017406	0c:f9:f3:18:00:05	0c:f9:f3:18:00:05	LOOP	60	Reply

На линках e2-e3 и e3-e3 между Switch-2 и Switch-3 у пакетов, MAC-адрес источника – Switch-2 (порты 2 и 3), Root BID – Switch-1 (Как нулевой порт).



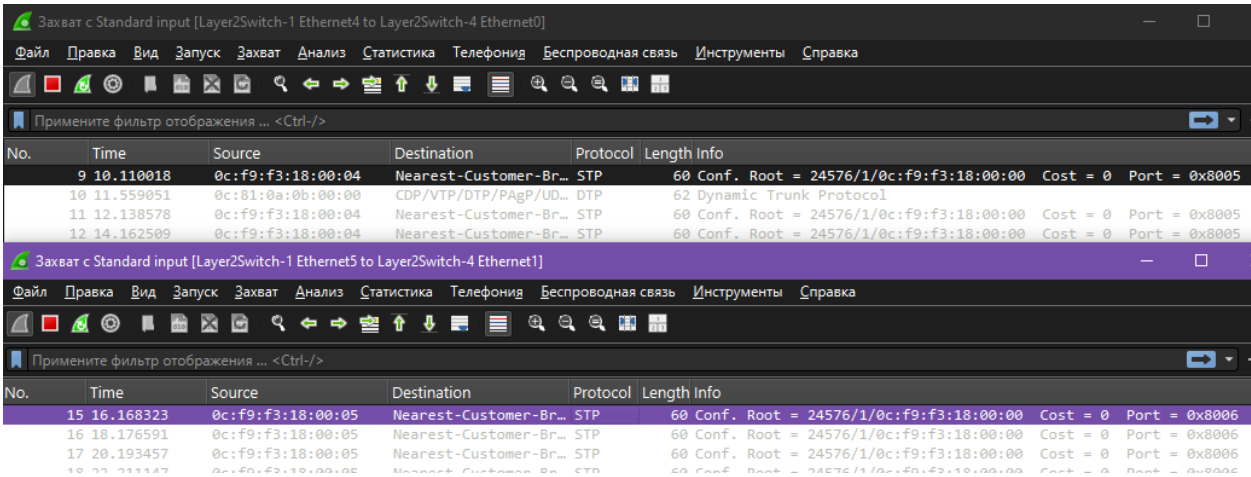
Захват с Standard input [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:9d:02:fd:00:02	0c:9d:02:fd:00:02	LOOP	60	Reply
2	0.940223	0c:9d:02:fd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8003
3	2.940037	0c:9d:02:fd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8003
4	4.939424	0c:9d:02:fd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8003

Захват с Standard input [Layer2Switch-3 Ethernet3 to Layer2Switch-2 Ethernet3]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:9d:02:fd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8004
2	1.999305	0c:9d:02:fd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8004
3	4.000137	0c:9d:02:fd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8004
4	4.898278	0c:77:66:b2:00:03	0c:77:66:b2:00:03	LOOP	60	Reply

На линках e4-e0 и e5-e1 между Switch-1 и Switch-4 у пакетов, MAC-адрес источника – Switch-1 (порты 4 и 5), Root BID – Switch-1 (Как нулевой порт).



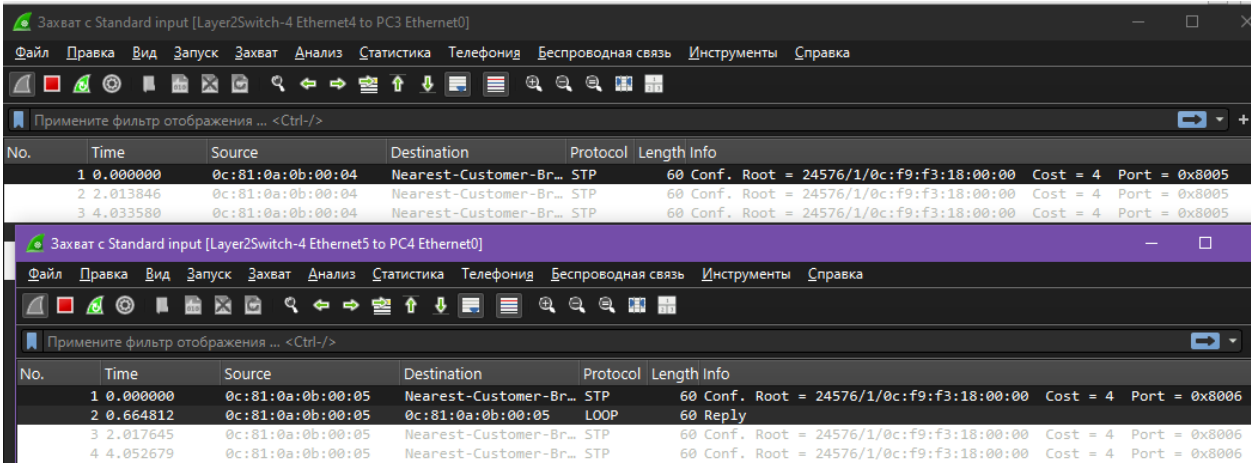
Захват с Standard input [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
9	10.110018	0c:f9:f3:18:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8005
10	11.559051	0c:81:0a:0b:00:00	CDP/VTP/DTP/PAGP/UD...	DTP	62	Dynamic Trunk Protocol
11	12.138578	0c:f9:f3:18:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8005
12	14.162509	0c:f9:f3:18:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8005

Захват с Standard input [Layer2Switch-1 Ethernet5 to Layer2Switch-4 Ethernet1]

No.	Time	Source	Destination	Protocol	Length	Info
15	16.168323	0c:f9:f3:18:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8006
16	18.176591	0c:f9:f3:18:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8006
17	20.193457	0c:f9:f3:18:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8006
18	22.211147	0c:f9:f3:18:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8006

На линках e4-e0 и e5-e0 между Switch-4 и PC3, PC4 у пакетов, MAC-адрес источника – Switch-4 (порты 4 и 5), Root BID – Switch-1 (Как нулевой порт).



Захват с Standard input [Layer2Switch-4 Ethernet4 to PC3 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:81:0a:0b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
2	2.013846	0c:81:0a:0b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
3	4.033580	0c:81:0a:0b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005

Захват с Standard input [Layer2Switch-4 Ethernet5 to PC4 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:81:0a:0b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
2	0.664812	0c:81:0a:0b:00:05	0c:81:0a:0b:00:05	LOOP	60	Reply
3	2.017645	0c:81:0a:0b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
4	4.052679	0c:81:0a:0b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006

На линках e4-e2 и e5-e3 между Switch-2 и Switch-4 у пакетов, MAC-адрес источника – Switch-2 (порты 4 и 5), Root BID – Switch-1 (Как нулевой порт).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:81:0a:0b:00:02	0c:81:0a:0b:00:02	LOOP	60	Reply
2	0.901889	0c:9d:02:fd:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
3	1.554959	0c:9d:02:fd:00:04	0c:9d:02:fd:00:04	LOOP	60	Reply
4	3.592876	0c:9d:02:fd:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
5	5.995896	0c:9d:02:fd:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:9d:02:fd:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
2	2.493142	0c:9d:02:fd:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
3	4.887833	0c:9d:02:fd:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
4	6.636458	0c:9d:02:fd:00:05	CDP/VTP/DTP/PagP/UD...	CDP	422	Device ID: vIOS-L2-01 Port ID: GigabitEthernet1/1
5	7.238594	0c:9d:02:fd:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006

На линках e6-e0 и e7-e1 между Switch-1 и Switch-5 у пакетов, MAC-адрес источника – Switch-1 (порты 6 и 7), Root BID – Switch-1 (Как нулевой порт).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:f9:f3:18:00:06	0c:f9:f3:18:00:06	LOOP	60	Reply
2	1.140007	0c:f9:f3:18:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8007
3	2.545713	0c:f9:f3:18:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8007
4	3.246017	0c:f9:f3:18:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8007
5	5.278021	0c:f9:f3:18:00:06	CDP/VTP/DTP/PagP/UD...	CDP	422	Device ID: vIOS-L2-01 Port ID: GigabitEthernet0/0

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:f9:f3:18:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8008
2	0.049061	0c:f9:f3:18:00:07	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol
3	2.111820	0c:f9:f3:18:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8008
4	3.046032	0c:f9:f3:18:00:07	0c:f9:f3:18:00:07	LOOP	60	Reply
5	4.185432	0c:f9:f3:18:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 0 Port = 0x8008
6	5.381020	0c:f9:f3:18:00:07	CDP/VTP/DTP/PagP/UD...	CDP	422	Device ID: vIOS-L2-01 Port ID: GigabitEthernet0/1

На линках e4-e0 и e5-e0 между Switch-5 и PC5, PC6 у пакетов, MAC-адрес источника – Switch-5 (порты 4 и 5), Root BID – Switch1 (Как нулевой порт).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:35:74:02:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
2	2.018436	0c:35:74:02:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
3	4.043483	0c:35:74:02:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005
4	6.071206	0c:35:74:02:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8005

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:35:74:02:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
2	2.025013	0c:35:74:02:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
3	4.052837	0c:35:74:02:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8006
4	6.089289	0c:35:74:02:00:05	0c:35:74:02:00:05	LOOP	60	Reply

На линках е6-е2 и е7-е3 между Switch-2 и Switch-5 у пакетов, MAC-адрес источника – Switch-2 (порты 6 и 7), Root BID – Switch1 (Как нулевой порт).

Захват с Standard input [Layer2Switch-2 Ethernet6 to Layer2Switch-5 Ethernet2]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:35:74:02:00:02	CDP/VTP/DTP/PAGP/UD...	DTP	62	Dynamic Trunk Protocol
2	0.797705	0c:9d:02:fd:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8007
3	2.826117	0c:9d:02:fd:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8007
4	4.845718	0c:9d:02:fd:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8007

Захват с Standard input [Layer2Switch-2 Ethernet7 to Layer2Switch-5 Ethernet3]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:9d:02:fd:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8008
2	2.019550	0c:9d:02:fd:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8008
3	3.842683	0c:9d:02:fd:00:07	0c:9d:02:fd:00:07	L2MP	60	Reply
4	4.057096	0c:9d:02:fd:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 24576/1/0c:f9:f3:18:00:00 Cost = 4 Port = 0x8008

То есть пакеты проходят по всем линкам, и даже при пересылке сохраняется информация о том, какой коммутатор является корневым.

5. Изменение стоимости маршрута для порта RP произвольно назначенного коммутатора и повторение действий из п.3;

Для коммутатора 4 изменил стоимость маршрута для порта RP с 12 условных единиц на 8.

Изменённая схема представлена в файле «lab2 5.png».

6. Сохранение файлов конфигурации устройств.

Файлы конфигурации устройств представлены в репозитории, в соответствии с названиями устройств.